

1 Dataset and EDA

Data: 10k images (28×28), 6 shape classes; exactly two non-zero counts per image; counts sum to 10.

Split: 9k train / 1k val.

Sanity checks: $\text{sum}==10$, exactly two non-zeros (no violations).

Distributions: class presence, count frequency, barplot of split into 135 classes (empty classes excluded).

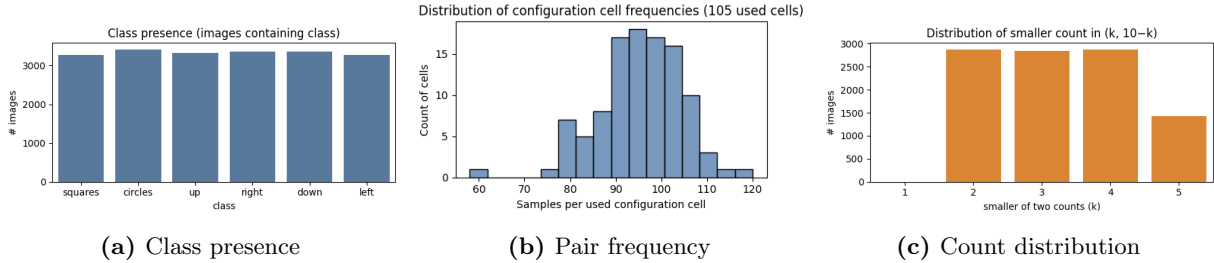


Figure 1: Dataset statistics.

EDA & Implications

Shape presence is balanced across the six classes. Distribution of smaller count shows that occurrences of counts 2-8 are also balanced. Counts 1 and 9 never appear, thus, only 105 of the 135 nominal configuration cells are populated. The per-configuration frequency barplot shows a dense central band (about 90–110 samples), a very short upper tail (few cells above 110), and a broader lower tail (many cells in 80–89). There is a pronounced gap separating the least-common outlier cell from the next least-common cell, indicating an isolated low-frequency outlier rather than a smoothly decaying tail. This represents moderate dispersion with one distinct low-frequency case.

Implications: no class or regression loss weighting; Macro-F1 reported to guard against configuration variance; watch out for elevated error in outlier configurations. Absence of (1, 9) splits may limit generalization to datasets containing such extremes.

Augmentations

We use geometric augmentations that preserve label semantics on binary images: random horizontal flip, vertical flip, and 90° rotations, while adjusting the direction of the occurring triangle shapes appropriately. This increases effective sample diversity without increasing the dataset size on disk.

We omit photometric augmentations (brightness/contrast, Gaussian noise) because the dataset is synthetic and pixel values are binary; such changes introduce grayscale artifacts that do not reflect the data-generating process and can blur edges. We also omit random erasing, which often removes parts of small shapes and corrupts the count labels.

2 Model

The network consists of a fixed shared backbone and two task-specific heads.

Backbone (shared feature extractor): A stack of four 2D convolution layers (channel progression $1 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64$) with 3×3 kernels, stride 1, padding 1 and ReLU activations, followed by flattening of the $64 \times 28 \times 28$ feature map and a fully connected layer to a 256-dimensional latent representation (ReLU). This backbone is unchanged across experiments.

Classification head: A $256 \rightarrow 256$ fully connected layer with ReLU, dropout (probability 0.3), then a $256 \rightarrow 135$ linear projection producing logits, converted to log-probabilities by a LogSoftmax layer, yielding $\log p \in \mathbb{R}^{135}$.

Regression head: A $256 \rightarrow 256$ fully connected layer with ReLU, dropout (probability 0.3), then a $256 \rightarrow 6$ linear layer producing raw continuous count estimates $\hat{\mathbf{c}} \in \mathbb{R}^6$.

Outputs: The forward pass returns $(\log p, \hat{\mathbf{c}})$ for each image.

Joint loss: Multitask training minimizes

$$\mathcal{L} = \underbrace{\text{NLLLoss}(\log p, y_{135})}_{\mathcal{L}_{\text{cls}}} + \lambda_{\text{cnt}} \cdot \underbrace{\text{SmoothL1Loss}(\hat{\mathbf{c}}, \mathbf{c})}_{\mathcal{L}_{\text{cnt}}}.$$

Setting $\lambda_{\text{cnt}} = 0$ yields classification-only; omitting $\mathcal{L}_{\text{cls}}$ yields regression-only. Dropout in each head regularizes the task-specific layers without altering shared feature extraction.

3 Results

Training was performed with Adam ($\text{lr} = 10^{-3}$), batch sizes 64 (train) and 1 000 (validation), for up to 100 epochs using early stopping on validation loss. The dataset split was fixed at 9 000 train / 1 000 validation samples. The global seed `torch.manual_seed(1)` was set for reproducibility.

Training was performed separately for three settings of the model:

- (1) **Classification-only:** $\lambda_{\text{cnt}} = 0$
- (2) **Regression-only:** $\mathcal{L} = \mathcal{L}_{\text{cnt}}$ (classification loss ignored)
- (3) **Multitask:** $\lambda_{\text{cnt}} = 2$

Setting	Val Loss	Top-1	Macro-F1	RMSE	MAE	Best epoch	Epochs	Time
Cls-only	1.2859	45.4%	33.2%	3.2073	1.8862	34	45	122.71s
Reg-only	0.1419	0.8%	0.1%	0.5831	0.3059	30	41	109.07s
Multitask	1.3835	52.5%	37.8%	0.5666	0.3089	30	41	111.68s

Table 1: Training metrics across the three settings.

Pair accuracy change	□	○	△	▷	▽	◁
□	–	+6.35	-2.85	+11.11	0.00	+1.45
○	+6.35	–	+4.00	-4.22	+4.92	0.00
△	-2.85	+4.00	–	0.00	+13.43	-3.08
▷	+11.11	-4.22	0.00	–	0.00	0.00
▽	0.00	+4.92	+13.43	0.00	–	0.00
◁	+1.45	0.00	-3.08	0.00	0.00	–

Table 2: Pairwise validation accuracies delta from classification-only to multitask setting.

Classification-only	□	○	△	▷	▽	◁
□	–	90.48%	95.71%	82.54%	93.75%	92.75%
○	90.48%	–	96.00%	98.59%	95.08%	95.31%
△	95.71%	96.00%	–	96.92%	86.57%	93.85%
▷	82.54%	98.59%	96.92%	–	91.94%	96.97%
▽	93.75%	95.08%	86.57%	91.94%	–	92.00%
◁	92.75%	95.31%	93.85%	96.97%	92.00%	–

Table 3: Validation accuracies for shape pairs, classification-only setting.

Multitask	□	○	△	▷	▽	◁
□	–	96.83%	92.86%	93.65%	93.75%	94.20%
○	96.83%	–	100.00%	94.37%	100.00%	95.31%
△	92.86%	100.00%	–	96.92%	100.00%	90.77%
▷	93.65%	94.37%	96.92%	–	91.94%	96.97%
▽	93.75%	100.00%	100.00%	91.94%	–	92.00%
◁	94.20%	95.31%	90.77%	96.97%	92.00%	–

Table 4: Validation accuracies for shape pairs, multitask setting.

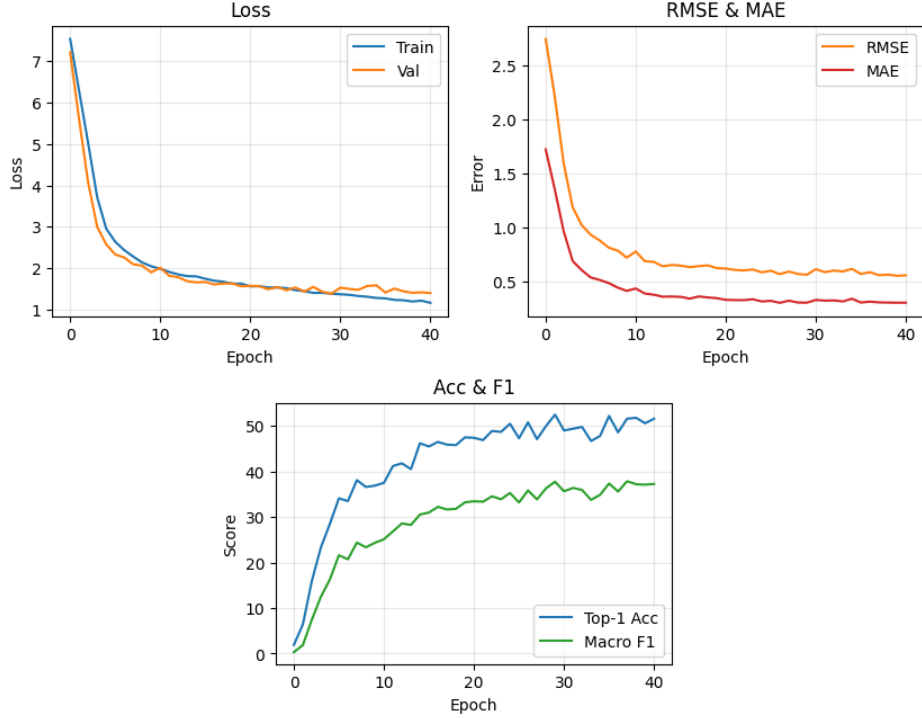


Figure 2: Training plots for multitask setting.

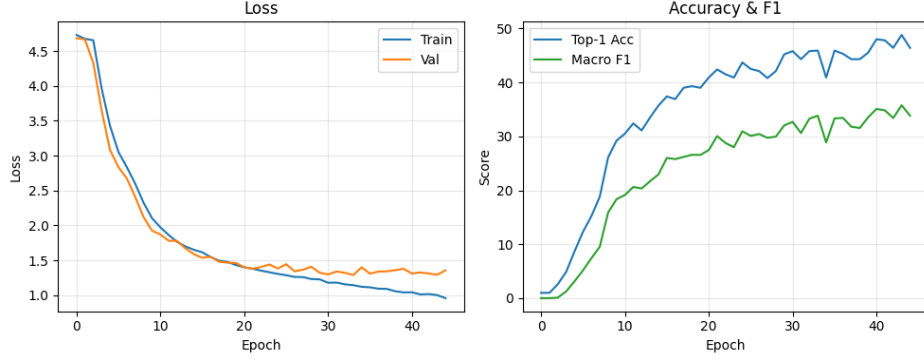


Figure 3: Training plots for classification-only setting

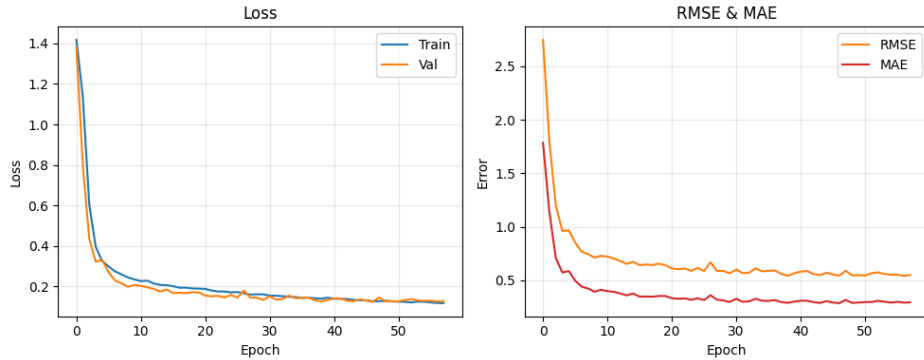


Figure 4: Training plots for regression-only setting.

4 Discussion

Multitask learning produced clear net positive transfer. Joint training increased validation Top-1 accuracy by 7.1 percentage points and Macro-F1 by 4.6 points over the classification-only baseline. Remarkably, the multitask model also improved RMSE relative to the regression-only configuration ($0.5831 \rightarrow 0.5666$), indicating that classification supervision helped suppress larger counting errors.

Pairwise shape analysis shows concentrated gains in challenging orientation or mixed-shape combinations (e.g. up-down +13.43 pp, square-right +11.11 pp). These improvements suggest the count head encourages the backbone to encode finer spatial/orientation cues that also benefit discrimination. A small number of pairs (square-up, circle-right, up-left) declined modestly (-3 to -4 pp), consistent with mild gradient interference across tasks. Some pairs remained unchanged (already near high performance), supporting that auxiliary regression acts primarily as a regularizer where headroom existed.

Training dynamics reinforce this interpretation: the multitask model reached its best epoch earlier and exhibited slower overfitting than classification-only, reflecting stabilizing shared supervision. The chosen $\lambda_{\text{cnt}} = 2$ balanced initial loss scales (classification $\approx 3.4\times$ regression), elevating regression’s influence enough to yield transfer without sacrificing primary classification accuracy.

Overall, gains in difficult orientation pairs and improved RMSE outweigh localized drops, making multitask learning well-suited for this dataset.